RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Friday, 25 September 2026, 8:33 AM
State	Finished
Completed on	Friday, 25 September 2026, 9:05 AM
Time taken	32 mins 17 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2		**g**	x	**t**	**x**	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>

int max(int a,int b){
    return (a>b)?a:b;
}

int lcs(char *s1,char *s2){
    int m=strlen(s1);
    int n=strlen(s2);
    int dp[m+1][n+1];
    for(int i=0;i<=m;i++){
        for(int j=0;j<=n;j++){
            if(i==0 || j==0){
                dp[i][j]=0;
            }
            else if(s1[i-1]==s2[j-1]){
                dp[i][j]=dp[i-1][j-1]+1;
            }
            else{
                dp[i][j]=max(dp[i-1][j],dp[i][j-1]);
            }
        }
    }
    return dp[m][n];
}
int main(){
    char s1[100],s2[100];
    scanf("%s",s1);
    scanf("%s",s2);
    printf("%d",lcs(s1,s2));
    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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